HoloDex: A Cost-Effective Open-Source Holonomic Mecanum Platform for Research and Education

Alisson Parizoto, Miguel M. R. Andrade, Héctor Azpúrua,
Douglas G. Macharet, and Paulo Rezeck
Computer Vision and Robotics Lab (VeRLab)
Department of Computer Science (DCC)
Universidade Federal de Minas Gerais (UFMG)
Belo Horizonte, Brazil

***Abstract*—Holonomic mobile bases offer a convenient setting for planar motion studies because they allow independent control of translational and rotational velocities without changing vehicle orientation. Research platforms that combine this mobility with mechanical rigidity under load, reproducible fabrication, and contemporary robotic middleware remain comparatively scarce in a cost-effective form. This paper presents *HoloDex*, an open-source four-wheel mecanum robot for research and education. The chassis uses aluminum structural profiles and 3D-printed mecanum wheels driven by stepper motors. A single microcontroller runs inverse and forward kinematics, motion supervision, wireless SoftAP teleoperation, and wheel odometry from magnetic shaft encoders, with an inertial measurement unit for yaw correction. Host-side ROS 2 connects through Eclipse Zenoh, so a high-performance computer need not ride on the robot. The paper describes the mechanical, electrical, and software design, formulates the kinematic model, and reports indoor motion trials with wheel-rate and encoder-aided odometry. The platform is completely open, allowing reproduction, modification, and use as a shared testbed for holonomic research and education.**

***Index Terms*—Holonomic robot, Mecanum wheels, Open-source hardware, Embedded control, Zenoh, ROS 2**

I. Introduction

Indoor mobile-robot experiments need bases that can translate and rotate in the plane and that connect to sensing, computation, and standard software stacks. This requirement is especially clear in navigation, perception, human-robot interaction, and multi-robot studies, where the mobile platform shapes both the protocol and its reproducibility [1]. Differential-drive robots remain common because they are mechanically simple to build and to control, yet their nonholonomic constraint forbids instantaneous lateral motion and often forces sequences of translation and reorientation. Holonomic platforms remove that restriction by offering three controllable planar degrees of freedom: longitudinal velocity, lateral velocity, and yaw rate [2]–[4]. Mecanum wheels realize this capability with a comparatively simple drivetrain, which explains their use in several educational and research prototypes [5], [6].

These needs create a practical trade-off. Commercial omnidirectional platforms offer stiff mechanics and mature software, but their price often limits fleets in teaching and research groups. Cost-effective and do-it-yourself builds reduce that barrier, yet they frequently rely on low-stiffness frames, ad hoc electronics, or software that does not connect easily to current robotic middleware. At the same time, computationally intensive algorithms need not reside on every mobile unit. In many experiments, the robot can run locomotion and odometry locally, while a workstation hosts perception, planning, and coordination. That split lowers cost, power, and maintenance per robot without leaving the ROS ecosystem.

This paper presents *HoloDex* (Fig. 1), an open-source four-wheel mecanum platform for research and education. Unlike low-cost educational platforms that rely on flexible acrylic or 3D-printed chassis and brushed DC gearmotors with substantial backlash, HoloDex features four distinct architectural advantages: (i) an aluminum-profile chassis that ensures structural bending stiffness under load; (ii) direct NEMA 23 hybrid stepper actuation providing high holding torque at low speeds ($\tau_h \approx 1.2\,\mathrm{N{\cdot}m}$), microstep determinism, and zero gearbox backlash; (iii) redundant shaft-side proprioception combining commanded microsteps with 12-bit magnetic absolute encoders and an IMU; and (iv) native ROS 2 integration offloaded to a host workstation through Eclipse Zenoh [7] over Wi-Fi, eliminating the power, weight, and financial cost of an onboard companion computer while preserving multi-hour battery autonomy.

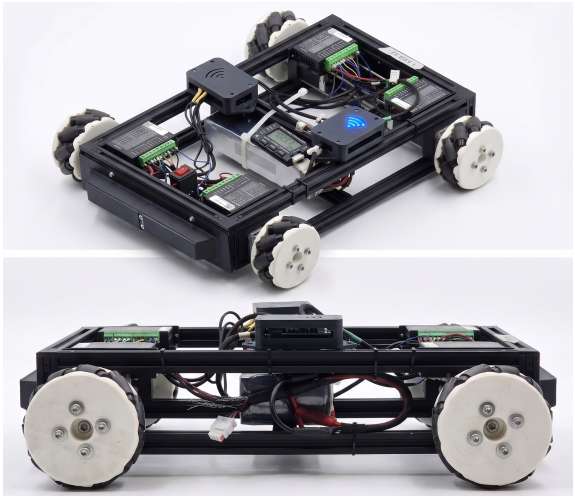


Fig. 1. HoloDex platform (top view): aluminum-profile chassis, four mecanum wheels with stepper drives, battery pack, and onboard control enclosures.

The contribution is an end-to-end holonomic base that brings together mechanics, embedded control, sensing, and middleware.[1] The paper does not introduce a new locomotion or estimation method. It documents an integrated platform and reports a first characterization of its motion. In particular, it (i) describes the mechanical and electrical architecture; (ii) formulates and implements the kinematic model; (iii) presents the firmware and the ROS 2 interface through Zenoh; and (iv) evaluates holonomic, rotational, and translational indoor motion using wheel-rate and encoder-aided odometry.

This study was financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior – Brasil (CAPES) – Finance Code 001, CNPq, and FAPEMIG. Corresponding author: rezeck@dcc.ufmg.br.

[1]The project repository is available at https://github.com/verlab/holodex.

II. Related Work

Holonomic wheeled locomotion spans several mechanism families that trade mechanical complexity, contact quality, and control effort. Pin and Killough [2] introduced orthogonal-wheel configurations with a formal kinematic treatment [8], and Holmberg and Khatib [3] developed powered-caster bases for mobile manipulation. Ball drives offer a further alternative when continuous ground contact is the priority [9]. In indoor service, mecanum and Swedish omni wheels remain common because independently driven hubs yield planar omnidirectional motion without active steering [5], [6]. Surveys and comparative studies review these architectures and their applications [5], [10], [11]. The ideal kinematic map, however, does not fully describe practice: roller-floor interaction introduces slip and systematic deviations from the nominal velocity mapping [4], [6]. That gap matters for the design and experimental characterization of a cost-effective research base.

Complete holonomic platforms span a wide range of cost, payload, and openness. Industrial systems such as the KUKA youBot [12] and Clearpath Ridgeback [13] combine integrated hardware, onboard computation, and mature ROS software, yet their price and proprietary construction hinder replication from commodity parts. Open designs occupy intermediate operating points. Robotont [14] offers a ROS-supported three-wheel omni platform near €2,000, whereas MoMo [15] pairs an aluminum-profile frame, geared motors, dual computers, and ROS 2 for payloads near $100\,\mathrm{kg}$ at substantially higher cost. LGDXRobot2 [16] likewise separates real-time motor control from an optional companion computer. Other platforms address more specific settings, including encoder-based estimation [17], robot-learning workloads [18], and outdoor localization [19].

At lower cost, Open Omnibot [20] and OmniRob [21] demonstrate microcontroller-driven mecanum platforms with wheel encoders. However, these designs prioritize minimal parts at the expense of structural rigidity (using laser-cut acrylic or 3D-printed plates prone to flex under payload), rely on brushed DC gearmotors with pronounced gearbox backlash and deadband, and lack native integration with ROS 2 middleware without a dedicated companion computer. Commercial kits, such as Nexus Robot 4WD systems [22]–[24], offer metal frames at moderate cost but ship with closed Arduino firmware and proprietary mechanics. HoloDex occupies a distinct position in this landscape: it combines the structural stiffness of an aluminum extrusion chassis and the precision of direct stepper actuation with host-offloaded ROS 2 interoperability via Zenoh, achieving a rigid, payload-capable research base under $1000 without requiring an onboard high-performance computer.

Table I compares representative platforms along drivetrain, frame, actuation, proprioceptive sensing, locomotion compute, middleware, payload, cost, and openness. HoloDex aims to combine structural rigidity, fabricability, embedded control, and ROS 2 interoperability, with an approximate bill of materials and open mechanical and firmware artifacts. A single microcontroller on an aluminum-profile, four-wheel mecanum chassis with stepper actuation performs inverse kinematics, motion supervision, teleoperation, and proprioceptive estimation. ROS 2 on the host reaches the robot through Eclipse Zenoh and therefore does not require an onboard companion computer.

III. System Design

This section presents the mechanical layout, kinematic model, electrical architecture, software stack, and proprioceptive estimators.

A. Mechanical Design

HoloDex uses a rectangular chassis of black anodized aluminum profiles (Fig. 1). Aluminum profiles offer bending stiffness under battery and sensor load, incremental reconfiguration, and wide part availability, unlike laser-cut acrylic. A transverse cross-member stiffens the frame and carries the control enclosures and power distribution. Corner modules mount four mecanum wheels with 3D-printed hubs and free rollers at $\pm 45^\circ$.

A NEMA 23 hybrid stepper drives each wheel. Steppers provide holding torque at low speed, deterministic microstepping, and compatibility with commodity chopper drivers (TB6600, up to $4\,\mathrm{A}$, DC 9–$42\,\mathrm{V}$), unlike brushed DC gearmotors. The nominal wheel radius is $r = 0.05\,\mathrm{m}$; the half-wheelbase and half-track in the kinematic model are $l_x = 0.19\,\mathrm{m}$ and $l_y = 0.195\,\mathrm{m}$ (axle and track spacings of $38\,\mathrm{cm}$ and $39\,\mathrm{cm}$). With 200 full steps per revolution and $32\times$ microstepping, the electrical resolution is $N_s = 6400$ steps/rev.

Fig. 2 shows the principal dimensions, motor placement, and mounting interfaces. Table II lists geometry, power, and conservative performance bounds used for inverse-kinematics saturation. Table III reports an approximate retail bill of materials.

Peak actuation follows from the motor holding torque $\tau_h \approx 1.2\,\mathrm{N{\cdot}m}$ and wheel radius $r = 0.05\,\mathrm{m}$, yielding a circumferential traction of $F_w = \tau_h / r = 24\,\mathrm{N}$ per wheel. For a four-wheel mecanum base with $\pm 45^\circ$ rollers, peak traction is direction-dependent. In pure longitudinal ($+x$) or lateral ($+y$) motion, all four wheels are driven; circumferential thrusts align along the commanded axis while opposing roller reactions cancel, giving an ideal peak force $F_{x,\max} = F_{y,\max} = 4(\tau_h / r) \approx 96\,\mathrm{N}$ (assuming sufficient floor friction μ). In diagonal translation ($\pm 45^\circ$), only two diagonally opposite wheels are powered while the remaining two roll passively, producing the minimum directional tractive envelope:

$$F_{\mathrm{diag,max}} = \frac{2\sqrt{2}\,\tau_h}{r} \approx 68\,\mathrm{N}. \tag{1}$$

For pure yaw about the chassis center, all four wheels generate tractive forces with moment arms $L = l_x + l_y = 0.385\,\mathrm{m}$, yielding an ideal peak yaw moment:

$$M_{z,\max} = \frac{4\,\tau_h\,L}{r} \approx 37\,\mathrm{N{\cdot}m}. \tag{2}$$

To estimate continuous acceleration and payload capacity, we consider the continuous-duty mecanum sizing relation along the worst-case diagonal direction: $m_{\max} = \frac{2\sqrt{2}\eta\tau_h}{ar}$, where $\eta \approx 0.5$ is a continuous-duty thermal derating factor ($\tau_{\mathrm{cont}} \approx 0.6\,\mathrm{N{\cdot}m}$) and $a \approx 1.0\,\mathrm{m/s^2}$ is the design acceleration on a flat indoor floor. This continuous tractive limit yields a total vehicle mass near $34\,\mathrm{kg}$ (or $\approx 68\,\mathrm{kg}$ in pure forward drive). Subtracting the measured curb weight of $9\,\mathrm{kg}$ (aluminum chassis, four NEMA 23 motors, drivers, and the $22000\,\mathrm{mAh}$ pack), the theoretical payload capacity is $\lesssim$ 25–$30\,\mathrm{kg}$. To maintain dynamic acceleration margins, prevent stepper microstep skipping during sudden velocity transients, and avoid mechanical compliance and fatigue in the 3D-printed hubs and couplers, the recommended payload is conservatively specified at $10\,\mathrm{kg}$ (safety factor ≈ 3). Experimental trials were conducted at the $9\,\mathrm{kg}$ curb weight carrying $\approx 3\,\mathrm{kg}$ of onboard instrumentation; destructive payload-to-stall validation was intentionally omitted to protect the 3D-printed assemblies. The firmware speed limits in Table II are likewise conservative relative to (1)–(2).

TABLE I
COMPARISON OF HOLONOMIC AND OMNIDIRECTIONAL PLATFORMS

Platform	Drive	Frame	Actuation	Sensing	Locom. compute	Middleware	Payload	Est. cost	Open
Orthogonal [2]	Orth. wheels	Custom	Custom	Custom	Custom	n.r.	n.r.	n.r.	Partial
Powered casters [3]	Casters	Custom	Servo/DC	Enc.	Workstation	n.r.	n.r.	n.r.	No
youBot [12]	4-mecanum	Industrial	Brushless	Enc.	Onboard	ROS	20 kg	≈€20,000	Softw.
Ridgeback [13]	Omni	Industrial	Brushless	Enc.	Onboard PC	ROS	≈100 kg	≈€48,000	Softw.
APR [17]	3-omni	Custom	DC	Wheel enc.	Onboard	Custom	n.r.	n.r.	Partial
Robotont [14]	3-omni	Sheet/3D	DC	Enc., IMU	Onboard	ROS	≤10 kg	≈€2,000	Yes
Open Omnibot [20]	4-mecanum	Acrylic/3D	Geared DC	Enc., IMU	ESP32	Web/app	n.r.	≈$173	Yes
OmniRob [21]	4-mecanum	Modular	Geared DC	Enc.	Arduino	n.r.	n.r.	≈$40	Yes
LGDXRobot2 [16]	4-mecanum	Acrylic	DC, PID	Enc., IMU	STM32, PC	ROS 2	n.r.	n.r.	Yes
MoMo [15]	4-mecanum	Al. profile	Geared DC	Enc.	NUC, PC	ROS 2	≈100 kg	≈€12,500	Yes
Nexus kit [22]	4-mecanum	Al. plate	DC, enc.	Opt. enc.	Arduino	ROS‡	≈20 kg	≈$700–$1400	No
TidyBot++ [18]	Casters	Custom	n.r.	Enc.	Onboard PC	ROS	n.r.	n.r.	Yes
HoloDex (ours)	**4-mecanum**	**Al. profile**	**Stepper**	**Enc., IMU**	**ESP32**	**ROS 2**	**≈10 kg**	**≈$500–$1000†**	**Yes**

Representative values from the cited publications and vendor listings. Payload and cost follow those sources; n.r. denotes a value not reported.
†Approximate retail BOM (Table III); excludes optional perception PC.
‡Community ROS packages; vendor kit ships with Arduino firmware.

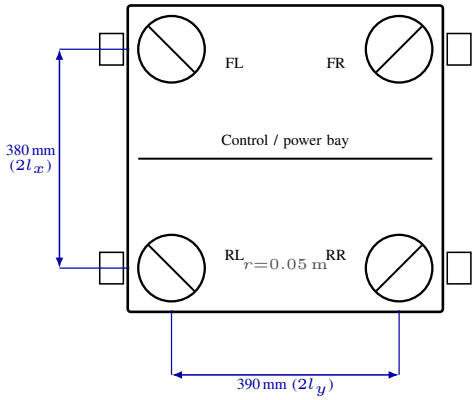


Fig. 2. Mechanical drawings of HoloDex: chassis dimensions, wheel/motor layout, and mounting interfaces for drivers and electronics.

TABLE II
HOLODEX PLATFORM SPECIFICATIONS

Parameter	Value
Drive type	4WD mecanum
Wheel radius r	0.05 m
Half-wheelbase l_x	0.19 m
Half-track l_y	0.195 m
Lever $L = l_x + l_y$	0.385 m
Steps / revolution N_s	6400
Stepper driver	TB6600, $\leq$ 4 A, DC 9–42 V
Battery	6S LiPo, 22.2 V nom., 22000 mAh (488 Wh)
Battery runtime (est.)	7–10 h continuous (> 12 h intermittent)
Logic supply	buck 5–30 V in / 5 V out, 3 A
Max. linear speed†	0.35 m/s
Max. yaw rate†	1.5 rad/s
Control loop / telemetry	50 Hz / 20 Hz
Comm. latency (Zenoh)	5–15 ms (802.11 Wi-Fi)
Holding torque (NEMA 23)	$\tau_h \approx 1.2$ N·m
Peak traction (diag / fwd)‡	≈ 68 N / ≈ 96 N
Peak yaw moment‡	≈ 37 N·m
Curb weight (measured)	9 kg
Tested payload margin	≈ 3 kg (instrumentation)
Payload (recommended / theor.)*	10 kg / $\lesssim$ 25–30 kg
Onboard compute (locomotion)	ESP32 (dual-core FreeRTOS)

†Firmware soft limits for teleoperation and command saturation.
‡Ideal peak from (1)–(2) at holding torque (friction/thermal derating not included).
*Payload capacity analytically estimated; experimentally evaluated up to tested payload.

B. Kinematic Modeling

The firmware converts a commanded body twist into four coordinated wheel rates and reconstructs that twist from shaft motion, so teleoperation and odometry share a single map. Body frame $\mathcal{B}$ sits at the geometric center of the wheel rectangle, with $+x$ forward and $+y$ to the left. The planar configuration is $\mathbf{q} = (x, y, \theta)^\top \in \mathrm{SE}(2)$, with position (x, y) of the origin of $\mathcal{B}$ and yaw θ. Instantaneous body velocity comprises longitudinal velocity v_x, lateral velocity v_y, and yaw rate ω, all in $\mathcal{B}$,

$$\mathbf{u} = \begin{bmatrix} v_x & v_y & \omega \end{bmatrix}^\top \in \mathbb{R}^3. \tag{3}$$

The hardware indexes the wheels as front-left (FL), rear-left (RL), front-right (FR), and rear-right (RR). With $L = l_x + l_y$, the inverse map from $\mathbf{u}$ to wheel linear speeds $\mathbf{v}_w = (v_{\mathrm{FL}}, v_{\mathrm{RL}}, v_{\mathrm{FR}}, v_{\mathrm{RR}})^\top$ is

$$\mathbf{v}_w = \begin{bmatrix} 1 & -1 & -L \\ 1 & 1 & -L \\ 1 & 1 & L \\ 1 & -1 & L \end{bmatrix} \mathbf{u} = \mathbf{J}\,\mathbf{u}, \tag{4}$$

where $\mathbf{J}$ is the inverse-kinematics allocation matrix. The mapping (4) produces pure yaw about the chassis center when $v_x = v_y = 0$ [5]. Signed step rates $\mathbf{h} = (h_{\mathrm{FL}}, h_{\mathrm{RL}}, h_{\mathrm{FR}}, h_{\mathrm{RR}})^\top$ (in hertz) follow

$$h_i = \frac{N_s}{2\pi r}\, v_{w,i}, \qquad i \in \{\mathrm{FL}, \mathrm{RL}, \mathrm{FR}, \mathrm{RR}\}, \tag{5}$$

where $v_{w,i}$ is the linear speed of wheel i. A left inverse of (4) yields the forward map

$$\mathbf{u} = \mathbf{J}^{+}\mathbf{v}_w, \qquad \mathbf{J}^{+} = \frac{1}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ -\frac{1}{L} & -\frac{1}{L} & \frac{1}{L} & \frac{1}{L} \end{bmatrix}, \tag{6}$$

which satisfies $\mathbf{J}^{+}\mathbf{J} = \mathbf{I}_3$, with $\mathbf{I}_3$ the 3×3 identity. World-frame rates of $\mathbf{q}$ follow

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{u}. \tag{7}$$

C. Electrical Architecture

Power distribution and sensing follow a modular star topology around a single ESP32 microcontroller (Fig. 3). A high-capacity 6S LiPo pack (22.2 V nominal, 22000 mAh, delivering 488.4 Wh) supplies the system through a master power switch, a 15 A inline protection fuse, and a panel voltmeter. The raw 22.2 V bus feeds the four TB6600 chopper drivers directly, operating well within their 9–42 V rating. A step-down buck converter regulates 22.2 V down to a stable 5.0 V

TABLE III
APPROXIMATE BILL OF MATERIALS FOR HOLODEX

Component	Qty	Est. unit / line
Aluminum profile and hardware	kit	$80–120
Mecanum wheels (3D-printed hubs/rollers)	4	$40–80
NEMA 23 hybrid steppers	4	$80–140
TB6600 stepper drivers (4 A)	4	$40–80
ESP32 DevKit and 3D-printed enclosures	1	$15–30
Buck Mini 3 A (5–30 V→5 V)	1	$3–8
PCA9548A and AS5600 modules	1, 4	$25–40
BMI160 (or equiv.) IMU	1	$5–15
6S LiPo 22.2 V / 22000 mAh and switch	1	$120–200
Misc. cabling, terminals, fasteners	set	$40–80
Approximate total BOM		**$450–775**‡

Retail estimates in USD; regional prices vary.
‡Lower bound assumes in-house printing and bulk sourcing; the $500–$1000 figure in Table I includes contingency, tooling consumables, and a conservative retail basket. Excludes optional host PC / Zenoh router.

(up to 3 A) for the ESP32 DevKit, which supplies 3.3 V to the sensor bus.

Total power consumption during continuous four-wheel locomotion ranges from 40 to 70 W (drawing ≈ 2.0–3.2 A at 22.2 V), while quiescent logic consumption is ≈ 1.5 W. Over the 488 Wh pack, this yields an estimated continuous operating runtime of 7–10 hours and > 12 hours under typical intermittent laboratory trials, avoiding mid-experiment battery swaps.

Dedicated GPIO lines drive each motor channel (STEP, DIR, and ENABLE). For sensing, a PCA9548A multiplexer isolates the four AS5600 12-bit magnetic encoders (channels 0, 1, 4, 5) sharing the default 0×36 I²C address, and connects a BMI160 6-DoF IMU on channel 6. The I²C bus operates in Fast-mode (400 kHz) with an 8 ms NACK timeout to guarantee deterministic sensor acquisition.

D. Software Architecture and Middleware

The software architecture partitions time-critical locomotion from asynchronous networking and host robotics middleware across four distinct layers (Fig. 4). The firmware treats the body twist (3) as the sole command that teleoperation and middleware clients may send, preserving (4)–(5) as the single source of truth for wheel coordination and acceleration limiting.

Firmware execution is assigned across the two cores of the ESP32 under FreeRTOS:

- **Core 1 (Real-Time Control Loop @ 50 Hz):** Runs the deterministic $\Delta t = 20$ ms locomotion loop. Each tick sweeps the PCA9548A multiplexer at 400 kHz, differentiates shaft angles, fuses rates via (8), applies inverse kinematics (4), enforces acceleration ramps, generates hardware pulses via FastAccelStepper, and integrates odometry states. A 600 ms watchdog stops the robot if commands lapse.
- **Core 0 (Asynchronous Network & Telemetry):** Manages the Wi-Fi radio stack, the SoftAP web server for teleoperation, and the Eclipse Zenoh [7] client. Odometry snapshots are captured from Core 1 using a non-blocking FreeRTOS mutex with zero timeout, guaranteeing telemetry publishing never stalls motor control. Inbound twists are latched via a lightweight spinlock.

Communication between robot and host uses 2.4 GHz 802.11 Wi-Fi via Eclipse Zenoh. Typical round-trip transport latencies range from 5 to 15 ms (sub-10 ms on dedicated routers), well within the 20 ms control period. Outbound

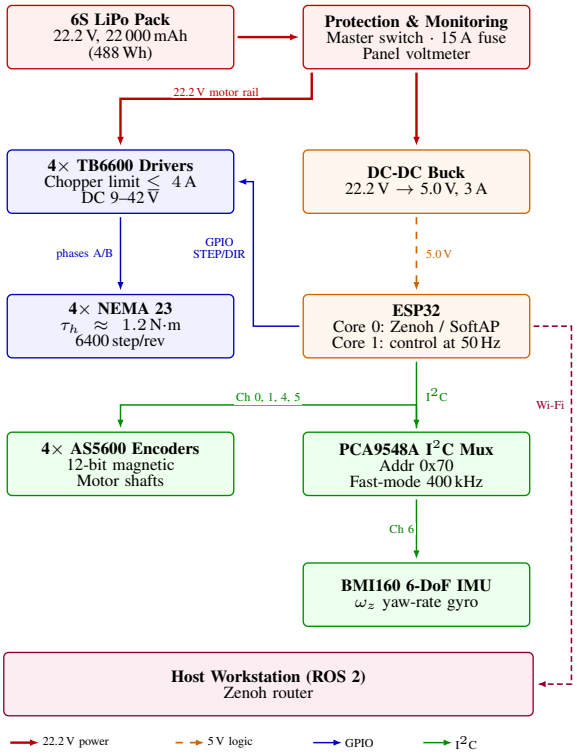


Fig. 3. Electrical architecture: 22.2 V pack power through switch and fuse to stepper drivers; buck-regulated 5.0 V logic rail; GPIO motor control; multiplexed 400 kHz I²C sensing; and 802.11 Wi-Fi link.

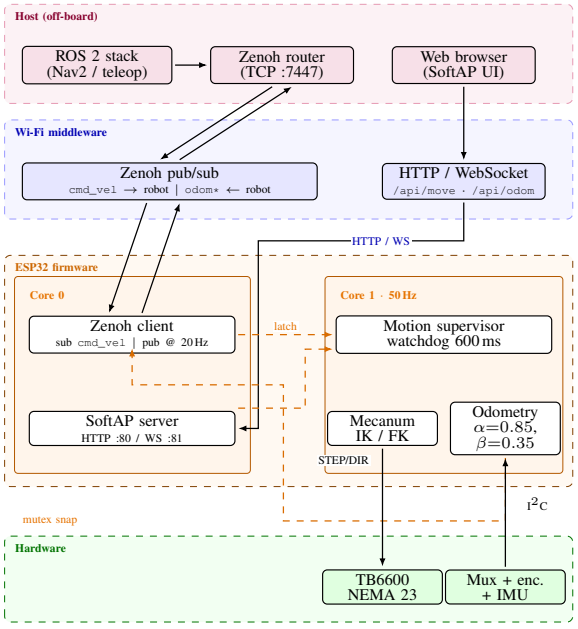


Fig. 4. Software architecture across execution layers: host ROS 2 and teleoperation; 802.11 Wi-Fi middleware; ESP32 dual-core FreeRTOS firmware; and physical hardware plant.

odometry telemetry is streamed at 20 Hz (50 ms). The workstation exposes standard ROS 2 topics through a Zenoh bridge without requiring an onboard companion computer.

A motion supervisor arbitrates SoftAP and Zenoh command sources under a timeout watchdog, applies acceleration-limited

wheel setpoints, and samples encoders and the IMU at a period $\Delta t = 20\,\mathrm{ms}$. SoftAP teleoperation needs no infrastructure Wi-Fi. When station mode is available, Eclipse Zenoh [7] carries structured messages between the robot and a host: inbound body-velocity commands and outbound streams for odometry, encoder rates, and IMU angular velocity. The platform uses ROS 2 on the workstation through Zenoh-compatible middleware; the microcontroller does not embed a ROS runtime.

E. Proprioception and Sensor Fusion

Two pose estimates are formed from the same motion and compared in Section IV. Logical step rates $\mathbf{h}^{\mathrm{fas}} \in \mathbb{R}^4$ come from the stepper trajectory generator (after motor polarity), and encoder rates $\mathbf{h}^{\mathrm{enc}}$ come from differentiating magnetic shaft-encoder angles and scaling by $N_s/(2\pi)$. When encoder samples are fresh, the estimator blends rates according to

$$h_i = \alpha\, h_i^{\mathrm{enc}} + (1-\alpha)\, h_i^{\mathrm{fas}}, \qquad \alpha \in [0,1], \tag{8}$$

where α is a blending weight (default $\alpha = 0.85$); otherwise $h_i = h_i^{\mathrm{fas}}$. The parameter $\alpha = 0.85$ prioritizes the physical shaft-measured rotation from the 12-bit magnetic encoders (providing direct observability of motor shaft displacement) while retaining a 15% smooth feedforward contribution from the microstep pulse generator to regularize high-frequency quantization noise caused by discrete angle differentiation at $50\,\mathrm{Hz}$ without introducing phase delay.

Converting $\mathbf{h}$ through the inverse of (5) and applying (6) yields a twist $\mathbf{u}^{\mathrm{w}}$. A first-order single-pole low-pass filter then smooths each component of $\mathbf{u}^{\mathrm{w}}$ at sample index k with smoothing factor $\beta = 0.35$,

$$\tilde{v}_{x,k} = \tilde{v}_{x,k-1} + \beta\big(v^{\mathrm{w}}_{x,k} - \tilde{v}_{x,k-1}\big), \tag{9}$$

and likewise v_y and ω, where $\tilde{v}_{x,k}$ is the filtered longitudinal velocity and $v^{\mathrm{w}}_{x,k}$ is the corresponding component of $\mathbf{u}^{\mathrm{w}}$. For the discrete sampling period $\Delta t = 20\,\mathrm{ms}$, $\beta = 0.35$ yields an effective continuous time constant $\tau = -\Delta t/\ln(1-\beta) \approx 46.4\,\mathrm{ms}$, corresponding to a $-3\,\mathrm{dB}$ cutoff frequency $f_c = 1/(2\pi\tau) \approx 3.4\,\mathrm{Hz}$. This bandwidth effectively suppresses stepper microstep detent ripple and differentiation chatter while preserving fast transient response ($< 100\,\mathrm{ms}$ rise time) during teleoperation and trajectory tracking.

The wheel-rate estimate follows the same chain from $\mathbf{h}^{\mathrm{fas}}$ alone. The encoder-aided estimate uses (8). World-frame pose follows the Euler discretization of (7). Both streams are shaft-side quantities: they track commanded microsteps and motor-shaft rotation, but they do not observe roller-floor slip or compliance in the printed hub and motor-wheel coupling. A BMI160 IMU supplies a yaw-rate measurement that the motion supervisor may use as a heading correction during lateral motion. Firmware includes a gyro-integrated pose, yet the reported trials omit it because bias drift produced heading estimates that diverged from the physical motion.

IV. Experimental Evaluation

To evaluate holonomic motion capabilities and examine the diagnostic consistency of shaft-side odometry, three complementary indoor protocols on a level floor isolate lateral strafing, pure in-place rotation, and extended linear translation. The three protocols decouple the primary planar degrees of freedom (v_x, v_y, ω) to test specific mecanum contact regimes:

1) **Holonomic square:** Traces a $1.2\,\mathrm{m}$ square at constant heading, sequencing forward, lateral strafe, reverse, and opposite-lateral translation, isolating roller-floor slip and heading drift during multi-directional holonomic motion.
2) **Translate-and-yaw squares:** Two successive squares formed by $1.2\,\mathrm{m}$ forward segments interleaved with eight in-place 90° turns, isolating pure rotational yaw kinematics about the chassis center.
3) **Orthogonal translation:** Out-and-back manual runs along surveyed laboratory axes of $2.3\,\mathrm{m}$, evaluating long-distance tracking consistency along decoupled cardinal directions.

Body-velocity commands followed (4) through the firmware path via Zenoh at the $9\,\mathrm{kg}$ curb weight. Both estimators start from a synchronized reset. Table IV reports the encoder-aided estimate, telemetry sample count, duration, return error e_{xy}, heading increment $\Delta\theta$, and planar RMSE relative to the wheel-rate estimate (RMSE_w).

TABLE IV
Encoder-Aided Odometry on Three Indoor Motion Protocols

Protocol	Dur. [s]	Samples	Path [m]	e_{xy} [m]	$\Delta\theta$ [deg]	RMSE_w [mm]
Holonomic square	38.8	726	5.20	0.30	16.3	7.8
Yaw squares	108.5	1949	10.42	0.18	727†	7.2
Orthogonal 2.3 m	51.4	958	9.34	0.12	89.0	11.7

RMSE_w: planar RMSE between the two shaft-side streams.
†Eight commanded 90° turns (720°). Telemetry logged at $\approx 18.7\,\mathrm{Hz}$ from the $50\,\mathrm{Hz}$ control loop.

A. Results

Across all trials, the two shaft-side streams remain in close agreement ($\mathrm{RMSE}_w = 7.2$–$11.7\,\mathrm{mm}$); what changes is how the reconstructed trajectory sits relative to intended ground marks (Fig. 5).

On the holonomic square (Fig. 5(a)), the estimated path length is $5.20\,\mathrm{m}$ against a geometric perimeter of $4.8\,\mathrm{m}$, yielding a return closure error of $e_{xy} = 0.30\,\mathrm{m}$ and residual heading of 16.3°. Physical displacement on the floor exhibited a matching return offset near $0.2\,\mathrm{m}$. The IMU yaw correction attenuated heading drift on the lateral legs without eliminating it, appearing as a progressive tilt in the reconstructed profile.

On the translate-and-yaw trial (Fig. 5(b)), forward segments remained close to $1.2\,\mathrm{m}$. Successive sides rotate slightly away from the axes ($e_{xy} = 0.18\,\mathrm{m}$). The integrated shaft heading reaches 727° after eight commanded right angles (720°), while high-rate rotation intervals account for approximately 76° per corner. Under pure yaw, opposing wheel velocities produce rotation about the geometric center, validating the physical mecanum kinematic allocation (4).

On the orthogonal translations (Fig. 5(c)), encoder-aided peak excursions reach $2.23\,\mathrm{m}$ and $2.36\,\mathrm{m}$ against surveyed floor marks of $2.3\,\mathrm{m}$. Net heading change after a manual 90° turn is 89.0°, with $e_{xy} = 0.12\,\mathrm{m}$ across two out-and-back runs, indicating that forward motion is the most repeatable regime.

B. Discussion

The close agreement between the wheel-rate and encoder-aided traces across all trials ($\mathrm{RMSE}_w = 7.2$–$11.7\,\mathrm{mm}$) demonstrates that the stepper actuation and microstepping pulse generation execute the commanded trajectory with high fidelity, confirming that the drives do not suffer from step skipping, stalls, or back-EMF slippage under nominal loads.

However, shaft-side agreement does not imply that the chassis follows the nominal kinematic path over ground. Both estimators operate strictly on the motor shaft; hence, external contact dynamics—including roller-floor slip, passive roller bearing play, hub wear, and torsional compliance in 3D-printed couplers—remain unobservable. With a $9\,\mathrm{kg}$ chassis, these contact phenomena become noticeable. 3D-printed hubs lack the strict concentricity of machined metal hubs, and compliant couplings allow angular windup under traction. In lateral motion, asymmetrical roller contact friction couples into

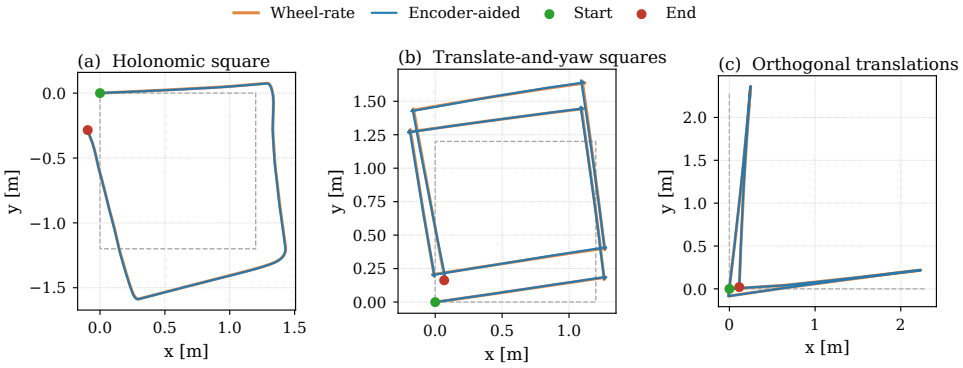


Fig. 5. Estimated planar trajectories from wheel-rate odometry and encoder-aided odometry. Dashed lines show the intended 1.2 m square in (a)–(b) and the surveyed 2.3 m axes in (c). Start and end markers refer to the encoder-aided estimate. Video of the experiments: https://youtu.be/orDW_grUFQE

parasitic yaw whenever the center of mass deviates from the geometric center, producing the drift in Fig. 5(a). Conversely, pure longitudinal motion avoids lateral roller scrubbing and yields the lowest return error (0.12 m).

The BMI160 gyro attenuated heading drift during strafing, but unmodeled bias drift precluded open-loop gyro integration. Two experimental limitations should be emphasized: (i) both estimators rely on proprioceptive shaft sensing and lack external ground-truth validation (such as an optical motion capture system) to quantify absolute tracking error and isolate dynamic slip; and (ii) the evaluation is based on characteristic benchmark runs for each canonical motion regime without multi-trial statistical averaging across diverse floor surfaces. Addressing these limitations with external ground-truth tracking and multi-trial statistical testing is central to future work.

V. Conclusion and Future Work

This paper presented *HoloDex*, an open-source four-wheel mecanum platform for research and education. HoloDex addresses the trade-off between structural rigidity, cost, and software integration by combining an aluminum-profile chassis, direct NEMA 23 stepper actuation, redundant shaft-side proprioception, and host-offloaded ROS 2 integration via Eclipse Zenoh without requiring an expensive onboard companion PC. Indoor motion trials across holonomic, rotational, and translational regimes show millimetre-level agreement between stepper and encoder estimators (7–12 mm RMSE), while highlighting the divergence between shaft-side odometry and ground displacement caused by unmodeled roller slip and coupler compliance. Future work will focus on integrating an optical motion capture system for external ground-truth benchmarking, collecting multi-trial statistical datasets across different floor surfaces, developing a high-fidelity simulation model, and fabricating CNC-machined aluminum wheel hubs and stiffer couplers. Subsequent research will also integrate onboard 2D/3D LiDAR sensing, mobile manipulation, and Nav2-based autonomous navigation under the unified body-velocity interface.

References

[1] B. Siciliano and O. Khatib, *Springer Handbook of Robotics*. Springer, 2 ed., 2016.

[2] F. G. Pin and S. M. Killough, "A new family of omnidirectional and holonomic wheeled platforms for mobile robots," *IEEE Transactions on Robotics and Automation*, vol. 10, no. 4, pp. 480–489, 1994.

[3] R. Holmberg and O. Khatib, "Development and control of a holonomic mobile robot for mobile manipulation tasks," *The International Journal of Robotics Research*, vol. 19, no. 11, pp. 1066–1074, 2000.

[4] J. A. Batlle and J. M. Font-Llagunes, "Holonomy in mobile robots," *Robotics and Autonomous Systems*, vol. 57, no. 4, pp. 433–440, 2009.

[5] S. G. Tzafestas, "Omnidirectional mobile robots: History, architecture and applications," *Journal of Intelligent & Robotic Systems*, 2018. Survey / review of omnidirectional platforms.

[6] L. Bruzzone and G. Quaglia, "Comparative analysis of mobile robot wheels design," in *International Conference on Robotics in Alpe-Adria Danube Region*, 2019.

[7] Eclipse Foundation, "Eclipse Zenoh: Zero overhead pub/sub, store/query and compute." https://zenoh.io, 2023.

[8] H. Asama *et al.*, "Design and traveling experiment of an omnidirectional holonomic mobile robot," in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 1992.

[9] M. Wada and H. Asada, "Design of a holonomic ball drive for mobile robots," in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 1999.

[10] F. Adăscăliţei and I. Doroftei, "Practical applications for mobile robots based on mecanum wheels—a systematic survey," *The Romanian Review Precision Mechanics, Optics & Mechatronics*, no. 40, 2011.

[11] P. S. Yadav, V. Agrawal, J. C. Mohanta, and M. F. Ahmed, "A theoretical review of mobile robot locomotion based on mecanum wheels," *Evergreen*, vol. 9, no. 2, pp. 387–397, 2022.

[12] R. Bischoff, U. Huggenberger, and E. Prassler, "KUKA youBot—a mobile manipulator for research and education," in *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 5544–5545, 2011.

[13] Clearpath Robotics, "Clearpath Ridgeback omnidirectional research robot." https://clearpathrobotics.com/ridgeback-indoor-robot-platform/, 2024. List price on the order of 48000 EUR as reported in comparative studies.

[14] T. Raudvere *et al.*, "ROBOTONT – open-source and ROS-supported omnidirectional mobile robot for education and research," *HardwareX*, 2023. Approx. hardware cost 2000 EUR; CERN-OHL / Apache 2.0.

[15] M. Knitt, M. Thakkar, Y. Elgohary, P. M. Braun, Y. Blank, and H. W. Rose, "MoMo: An open-source modular mobile robot research platform," in *Proceedings of the XXV International Conference on Material Handling, Constructions and Logistics (MHCL)*, 2024. Approx. parts budget 12500 EUR; payload about 100 kg; ROS 2.

[16] LGDXRobotics, "LGDXRobot2: Build a ROS 2 robot with decent performance." https://www.hackster.io/lgdxrobotics/lgdxrobot2-build-a-ros-2-robot-with-decent-performance-e082fe, 2024. STM32 real-time layer + optional onboard PC; mecanum.

[17] J. Morín *et al.*, "Design, implementation and validation of the three-wheel holonomic motion system of the assistive personal robot (APR)," *Sensors*, vol. 16, no. 10, p. 1658, 2016.

[18] J. Wu *et al.*, "TidyBot++: An open-source holonomic mobile manipulator for robot learning," in *Conference on Robot Learning (CoRL)*, 2024. Open-source holonomic mobile base.

[19] B. McGuinness *et al.*, "Mechatronic design and development of a non-holonomic omnidirectional mobile robot for automation of primary production," *Cogent Engineering*, 2017. MARIO platform with multi-sensor localization.

[20] S. Hoque, "Open omnibot: An open-source mecanum-wheel mobile robot platform." https://github.com/samiul-hoque/open-omnibot, 2025. BOM approximately USD 173; ESP32, encoders, BNO055.

[21] C. Xiang, "OmniRob: Low-cost modular mecanum robot." https://github.com/CassiusXiang/OmniRob, 2023. BOM approximately USD 40; Arduino-based.

[22] NEXUS Robot, "NEXUS robot: Mecanum and omni wheel platforms." https://www.nexusrobot.com/, 2024.

[23] Amicus Engineering, "4wd mecanum wheel mobile robot kit." https://amicus.com.sg/products/4wd-mecanum-wheel-mobile-robot-kit/, 2024. Nexus-based kit listing; approx. SGD 1350.

[24] KTH Smart Mobility Lab, "SML nexus 4wd holonomic robots." https://github.com/KTH-SML/sml_nexus, 2021. University customization of Nexus mecanum bases with ROS.